AUTONOMOUS DENGUE BREEDING SPOT DETECTOR ROBOT

- GROUP 1 MEMBERS:
  - IFTEQUR AHMED (22299172)
  - SHAFI AHMED ADIB (23101307)
  - SADAT HOSSAIN (22201314)
  - FATIN RAHMAN TASEEN (22299002)
  - MD. NAFIZUL SADIK (21201313)

CSE461: INTRODUCTION TO ROBOTICS

SECTION 4 | INSTRUCTOR: UTSHA KUMAR ROY AND SHREYA ADRITA BANIK

ABSTRACT

This project presents an autonomous robotic system designed to combat the dengue crisis by detecting stagnant water breeding grounds for Aedes mosquitoes. The 2WD robot navigates rooftops and courtyards using a systematic grid patrol strategy aided by wheel encoders and ultrasonic obstacle avoidance. To maximize detection accuracy and minimize false positives, it utilizes a sensor fusion approach, combining a moisture sensor, IR sensor, and DHT11 humidity sensor through a voting logic algorithm. Upon detecting potential hazards, a servo-mounted mechanism tilts the sensor array for a confirmatory secondary scan before triggering an audiovisual alert. This low-cost, automated solution offers a scalable method for continuous surveillance, significantly improving urban public health management.

INTRODUCTION

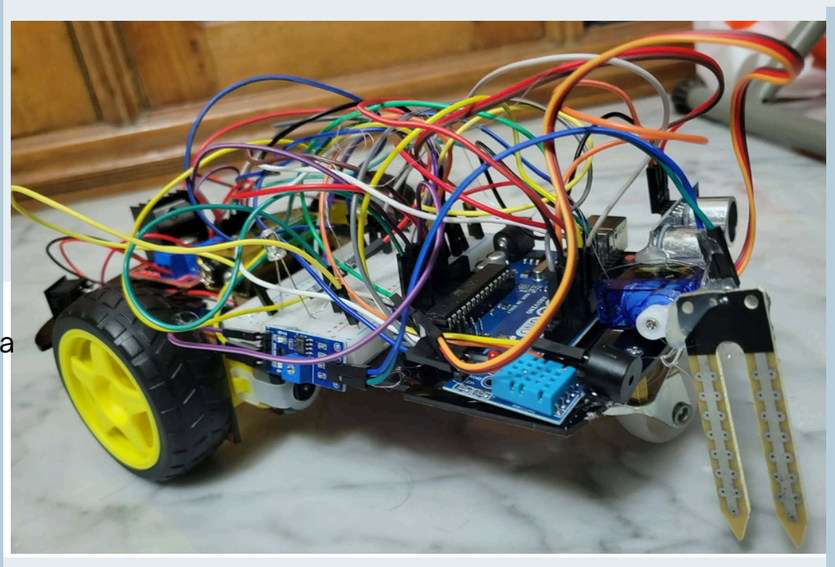
- Background & Motivation: Bangladesh faces severe, recurring dengue outbreaks every monsoon season, with thousands of hospitalizations annually.
- Problem Statement: The majority of Aedes mosquito breeding spots (stagnant water pools) on rooftops and construction sites go unnoticed due to a lack of regular manual inspection.
- Objectives: To build a low cost, automated robot that continuously patrols outdoor areas, scans for stagnant water, confirms findings using multi-sensor logic, and alerts residents to the hazard.
- Real-world Relevance: Provides a scalable early-warning system to prevent mosquito infestations before they develop, directly contributing to public health improvement in densely populated areas.

HARDWARE COMPONENTS

- Arduino Uno: Central logic and microcontroller.
- L298N Motor Driver: High-current power delivery to the wheels.
- Breadboard, Wires, Resistor, LED, Buzzer
- HC-SR04 Sonar: Ultrasonic distance measurement.
- DHT11 Temperature And Humidity Sensor Module
- YL-69 Soil Hygrometer Humidity & Soil Moisture Detection Sensor
- Li-ion Battery

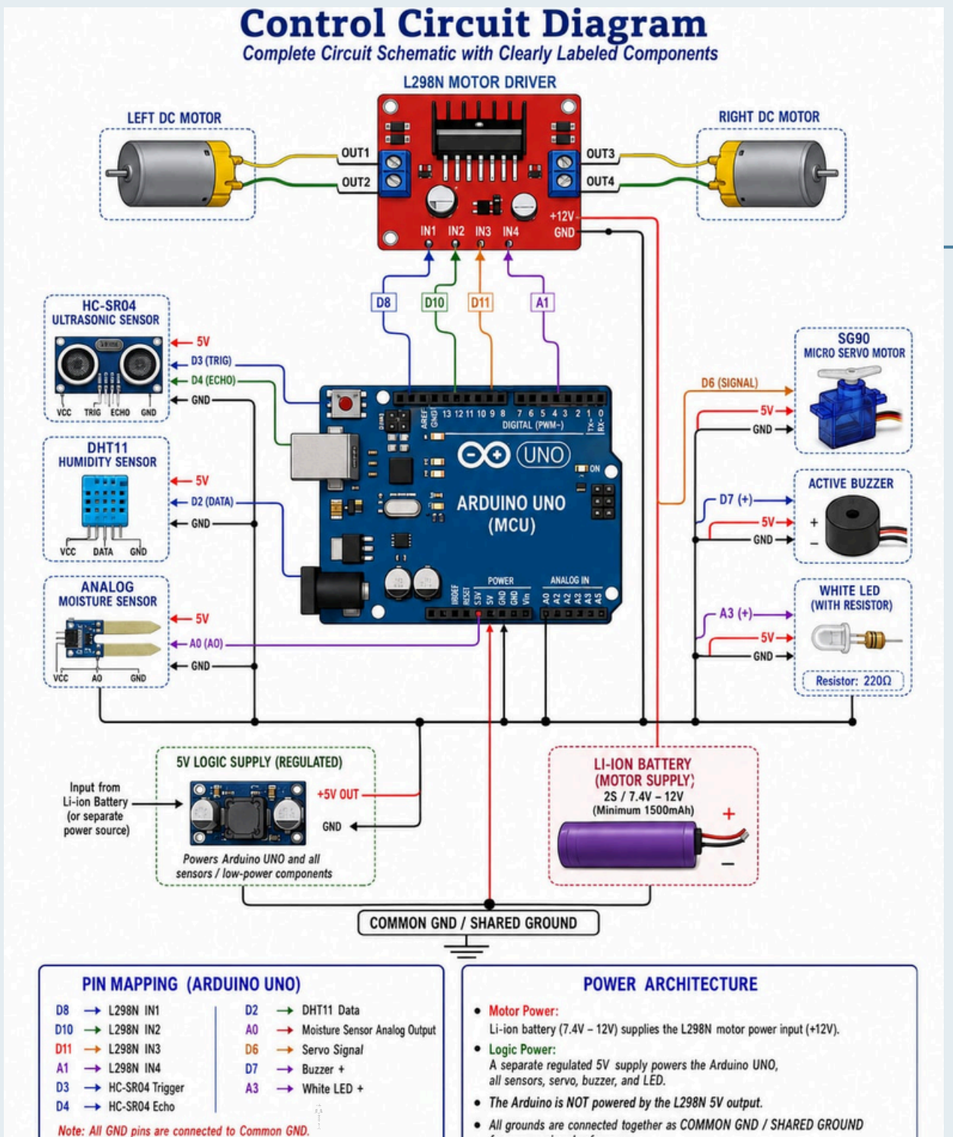
FUNCTIONALITIES

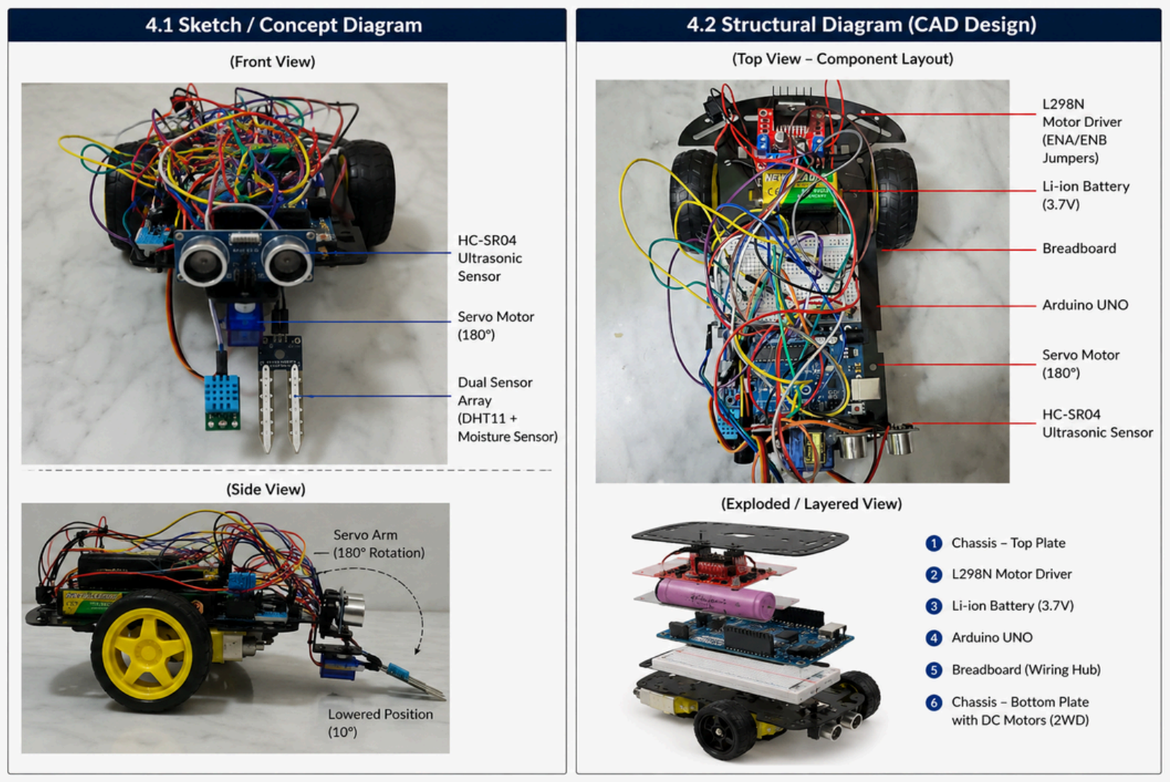
- 6.1 Mechanical Functionality: Operates on a differential drive 2WD system for mobility. It utilizes a dedicated servo motor as a robotic "neck" to physically tilt the scanning module toward the ground for close range container inspection.
- 6.2 Software Functionality: Implements a "voting logic" algorithm requiring at least two out of three sensors (Moisture, IR, Humidity) to trigger. Includes a 3-second pause-and-confirm stage to drastically reduce false positives.
- 6.3 Navigation Functionality: Uses infrared wheel encoders to maintain a systematic grid/zig-zag patrol path, relying on the ultrasonic sensor to reverse and turn away from obstacles within a 20cm threshold.
- 6.4 Software-Hardware Communication: The Arduino continuously reads analog/digital data from the sensor array and outputs precise PWM (Pulse Width Modulation) signals to the L298N for motor actuation.



APPLICATIONS AND COMPETITORS

- Real-world Applications: Urban sanitation patrols, warehouse monitoring, and residential mosquito control.
- Comparison / Competitors: Competes with aerial mapping drones (expensive, difficult indoors) and manual human inspection.
- Advantages: Highly cost-effective, fully autonomous, and features zero power-drop restarts. Voting logic eliminates false positives.
- Limitations: Navigation physics are currently restricted to smooth, flat indoor surfaces.





CONCLUSION

- Summary: Successfully prototyped an autonomous robot capable of navigating a grid, identifying stagnant water via sensor fusion, and marking the hazard for human intervention.
- Key Achievements: Integrated multi-sensor voting logic to prevent false alarms and implemented an active servo-targeting system for accurate ground scanning.
- Future Improvements: Integrating an SD card module to log hazard locations with timestamps, and adding a physical marking mechanism (e.g., a secondary servo to drop chalk powder on the detected spot).